

Four-Wave Mixing in Hot ^{85}Rb : A Physics-First Walkthrough of the GABES Engine

Current FWM Physics Model — revised 2026-07-03
(first-principles coupling, collisional decoherence, corrected IDS formula,
hardened Ultra noise, predictive biphoton)

Personal study notes

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Abstract

These notes derive, step by step, every piece of physics that the GABES four-wave-mixing (FWM) code evaluates — from the open-system master equation of the ^{85}Rb D_1 double- Λ system, through the Doppler average and the truncated-Floquet sideband steady state, to the Maxwell–Bloch propagation that turns the atomic susceptibility into seed/conjugate gain, intensity-difference squeezing, and (in the second mode) a spontaneous-FWM biphoton waveform. The macroscopic coupling is now first-principles: the hyperfine Clebsch–Gordan structure inside $\bar{\chi}$ is completed by the explicit population/degeneracy normalization $p_F/[2(2I+1)]$ and the pure- ^{85}Rb CRC vapour density, leaving only a dimensionless residual line-strength calibration (≈ 1). The dissipator carries density- and temperature-dependent collisional decoherence (optical self-broadening $\Gamma + \beta N$ and ground-state Raman collisions), so the achievable squeezing peaks near 121–131 °C and degrades at higher temperature, as observed. The ideal intensity-difference noise uses the conserved-difference Bogoliubov result $S_{\text{ideal}} = (G_s - G_c)^2/(G_s + G_c)$ [= $1/(2G_s - 1)$ in the lossless limit], replacing an older expression that overestimated squeezing at high gain. Seeded squeezing includes the reference 0.32° pump–probe longitudinal mismatch from Balanced fidelity up; High fidelity adds a conservative χ -reused refractive correction and segmented propagation; Ultra adds 64-segment dynamic pump-budget depletion, Gaussian beam overlap, in-cell loss/noise, Zeeman diagnostics, and — intrinsic since the v5 hardening — an excess-noise model with arm-resolved linear absorption of probe and conjugate plus OD-anchored pump scatter. Biphoton mode defaults to a predictive first-principles Doppler-averaged waveform (Autler–Townes two-photon denominator, velocity-class coherent sum on an auto-refined grid, natural-linewidth decay) and uses vector phase matching: calibrated longitudinal Δk_z plus absolute transverse Δk_x , with a two-dimensional signal–idler vector acceptance map in Fine mode. The heavy OBE grid can be evaluated by optional compiled Numba kernels; the (Δ, T) squeezing scan builds one master $\bar{\chi}(\delta, \Delta_{\text{eff}})$ table *per temperature* (the collisional dissipator makes $\bar{\chi}$ T -dependent). The emphasis is on the *physics*; the code is referenced only as a map so that each equation can be located in `gabes/`. The direct `compute_spectrum` path is regression-anchored to G. Sim, H. Kim, and H. S. Moon, *Sci. Rep.* **15**, 7727 (2025).

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1 Purpose and the object we compute

Non-degenerate four-wave mixing in a hot alkali vapour is a third-order parametric process. A strong *pump* drives a multilevel atom; a weak *seed* (probe) injected on one Raman leg stimulates the emission of a *conjugate* on the other leg. Energy and momentum conservation tie the two generated photons together, so signal and conjugate emerge as a two-mode squeezed (twin-beam) state: their *intensity difference* is quieter than the shot-noise limit, even though each beam alone is bright and thermal-like.

The code answers two questions:

1. **Seeded / squeezing mode.** Given a one-photon detuning Δ , cell temperature T , pump power, seed power, and detection efficiency, what is the seed power gain G_s , the conjugate gain G_c , and the intensity-difference squeezing S (in dB) as a function of the two-photon detuning δ ? This is the physics of Sim *et al.*, and is the regression-anchored core.
2. **Spontaneous / biphoton mode.** For a cascade or diamond topology with *no* seed, what is the time-correlation $g_{SI}^{(2)}(\tau)$ of the spontaneously generated signal-idler pair, its width, pair rate, and coincidence-to-accidental ratio (CAR)? The default Predictive model computes the

waveform, width and bandwidth from first principles, with a reference-anchored absolute rate; a legacy Calibrated model is kept for comparison. Either way it is a source estimate, not a full quantum-Langevin propagation.

The whole numerical machine exists to evaluate one object accurately: the *atomic susceptibility matrix* $\bar{\chi}(\delta, \Delta; v, T)$ that links the induced optical polarizations on the two Raman legs to the seed and conjugate fields. Everything before it (Hamiltonian, dissipator, Floquet, Doppler) builds that matrix; everything after it (Maxwell–Bloch, saturation, squeezing formula) reads it out.

Code map. The seeded pipeline lives in `gabes/schemes/fwm.py::compute_spectrum`; the physics-agnostic OBE engine is `gabes/core.py`; the susceptibility $\rightarrow$ observable layer is `gabes/observables.py`; the level model and rates are `gabes/atoms.py` and `gabes/hyperfine.py`; the thermal average is `gabes/doppler.py`; physical constants and field helpers are `gabes/constants.py`. The biphoton mode is the `GenericFWMSolver` class in the same `fwm.py`.

2 The double- Λ system and its energy scales

2.1 Levels

The seeded model uses the ^{85}Rb D_1 line ($5S_{1/2} \rightarrow 5P_{1/2}$, $\lambda \approx 795\text{ nm}$) reduced to four levels (`atoms.double_lambda_rb85`; the builder now takes the dephasing rates γ_{gg} and γ_{opt} as arguments, see Sec. 4.3):

index	state	manifold	role
g_1	$ 5S_{1/2}, F=2\rangle$	ground	lower Raman state
g_2	$ 5S_{1/2}, F=3\rangle$	ground	upper Raman state
e_2	$ 5P_{1/2}, F'=2\rangle$	excited	optical
e_3	$ 5P_{1/2}, F'=3\rangle$	excited	optical

The two ground hyperfine states are split by $\nu_{\text{HF}} = \nu_{\text{ground HF}} = 3.0357\text{ GHz}$; the two excited states by $\nu_{\text{HF}}^{(e)} = 361.58\text{ MHz}$. The geometry is “double- Λ ”: each ground state couples optically to *both* excited states, and the two ground states share the excited manifold. That shared manifold is what lets a seed on the $g_2 \leftrightarrow e$ leg transfer coherence to the $g_1 \leftrightarrow e$ leg and generate the conjugate — the four “waves” are pump, pump, seed, conjugate.

2.2 Hyperfine line strengths (Clebsch–Gordan)

Each $F \rightarrow F'$ link carries a relative strength C_F^2 from the Wigner $3j/6j$ sum (`hyperfine.CF2`), exact rationals for ^{85}Rb D_1 :

$$C_F^2(2, 2) = \frac{10}{81}, \quad C_F^2(2, 3) = \frac{35}{81}, \quad C_F^2(3, 2) = \frac{35}{81}, \quad C_F^2(3, 3) = \frac{28}{81}. \quad (1)$$

These factors enter the physics in three distinct places, and using one consistent quantity for all three is what makes the model self-consistent:

1. **Coherent drive.** The effective Rabi frequency on link $g \rightarrow e$ is scaled by $\sqrt{3 C_F^2(F_g, F_e)}$ (`TRANSITION_DIPOLE_SCALE`); the factor 3 is the $J \rightarrow J'$ reduction of the D_1 reduced matrix element.

2. **Polarization readout.** The macroscopic polarization on a leg is the same CG-weighted sum of optical coherences (Sec. 7).
3. **Spontaneous emission branching.** An excited state e decays to ground g at the branched rate

$$\Gamma_{e \rightarrow g} = \Gamma \frac{C_F^2(F_g, F_e)}{\sum_{g'} C_F^2(F_{g'}, F_e)}, \quad \sum_g \Gamma_{e \rightarrow g} = \Gamma, \quad (2)$$

so total population leaves each excited state at the natural rate Γ (`atoms.double_lambda_rb85`).

2.3 Energy scales and why the rotating frame is mandatory

The relevant frequencies span fifteen orders of magnitude: the optical carrier is $\omega_{D_1}/2\pi = 377$ THz, the hyperfine splitting 3 GHz, the natural linewidth $\Gamma/2\pi = 5.746$ MHz, the ground-coherence decay 100 kHz. No solver integrates the bare optical oscillation. Instead we move to a frame co-rotating with the pump and apply the rotating-wave approximation (RWA): all optical carriers are removed, leaving *detunings* of order MHz–GHz on the diagonal and slowly varying Rabi couplings off-diagonal. Throughout, energies are expressed as angular frequencies (i.e. $\hbar \equiv 1$ in the Hamiltonian), so “ H ” below is really $H/\hbar$ in rad s^{-1} .

3 Units, fields, and the vapour density

3.1 Rabi frequency from beam power

A beam of power P and $1/e^2$ intensity radius w_0 has peak intensity $I = 2P/(\pi w_0^2)$. The code converts this to a Rabi frequency through the saturation intensity (`constants.rabi_freq`):

$$I = \frac{2P}{\pi w_0^2}, \quad \Omega = \Gamma \sqrt{\frac{I}{2I_{\text{sat}}}}, \quad I_{\text{sat}} = 4.484 \text{ mW cm}^{-2}. \quad (3)$$

Equation (3) is just the standard two-level relation $I/I_{\text{sat}} = 2\Omega^2/\Gamma^2$ solved for Ω ; tying Ω to I_{sat} keeps the drive strength anchored to a measured quantity rather than to an absolute dipole. With the paper geometry ($w_{\text{pump}} = 530 \mu\text{m}$, $w_{\text{seed}} = 330 \mu\text{m}$, both $1/e^2$ radii), 600 mW of pump gives $\Omega_{\text{pump}}/2\pi$ of order a GHz, while 8 μW of seed gives $\Omega_{\text{seed}}/2\pi$ of order MHz — a clean separation of “strong pump, weak seed” that the linear-response treatment below relies on.

3.2 Number density

The atomic density sets the optical depth and hence the gain. The seeded model uses the *pure-⁸⁵Rb* CRC vapour-pressure fit (`hyperfine.number_density`, isotopic fraction $f_{85} = 1$) — the same density that lets the validated absorption scheme reproduce the lab AutoOD transmission to $< 0.1\%$:

$$\log_{10} P_{\text{Pa}} = \begin{cases} 9.863 - 4215/T, & T < 312.45 \text{ K (solid Rb)}, \\ 9.318 - 4040/T, & T \geq 312.45 \text{ K (liquid Rb)}, \end{cases} \quad N = \frac{P}{k_B T}. \quad (4)$$

At the reference 121 °C this gives $N = 2.15 \times 10^{19} \text{ m}^{-3}$. The enriched cell of the reference experiment is very nearly pure ⁸⁵Rb, so the previously used natural-abundance Steck fit (`atoms.rb85_density`, $f_{85} = 0.7217$; retained for the other schemes) understated N for the FWM cell by roughly the abundance factor. The exponential temperature dependence of P is the origin of the model’s extreme sensitivity to T : a few degrees changes N by tens of percent, and the gain is exponential in N (Sec. 9). The biphoton mode uses the per-isotope elemental densities in `species.number_density` (CRC fit for Rb, Steck for Cs).

4 Open-system dynamics: the master equation

The atom is open: it spontaneously emits, and ground coherences dephase from collisions and transit. The state is therefore a density matrix ρ obeying a Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation,

$$\dot{\rho} = -i[H, \rho] + \mathcal{D}[\rho], \quad (5)$$

with H the rotating-frame Hamiltonian (Sec. 4.1) and $\mathcal{D}$ the dissipator (Sec. 4.2). We only ever want the *steady state* ($\dot{\rho} = 0$), since the experiment is continuous-wave.

4.1 The Hamiltonian

In the rotating frame the diagonal of H carries detunings and hyperfine splittings, and the off-diagonal carries the (RWA) drives. The static part at zero effective detuning is built by `static_hamiltonian_at_Deff_zero`:

$$H_0 = \underbrace{\delta |g_2\rangle\langle g_2| - \omega_{\text{HF}}^{(e)} |e_2\rangle\langle e_2|}_{\text{diagonal: two-photon \& excited HF}} + \underbrace{\sum_{\text{legs}} \frac{\Omega_{ge}}{2} (|g\rangle\langle e| + |e\rangle\langle g|)}_{\text{RWA drives}}, \quad (6)$$

where δ is the two-photon (Raman) detuning, $\omega_{\text{HF}}^{(e)} = 2\pi\nu_{\text{HF}}^{(e)}$ places e_3 at zero and e_2 below it, and the leg drive is $\Omega_{ge} = \Omega\sqrt{3C_F^2(F_g, F_e)}$ from Eq. (1). The one-photon detuning Δ and the atomic velocity do *not* appear in H_0 ; they enter later as a single shift Δ_{eff} on the excited diagonal (Sec. 5), which is the trick that makes the Doppler average cheap.

Which field sits where (the branch). A subtlety unique to FWM: the pump and the weak fields can be assigned to the two legs in two inequivalent ways, corresponding to the two Raman resonances. The code labels them `branch =  $\mp 1$` . For the standard red-detuned seed (branch -1), the pump Ω_A drives the g_1 leg and the seed Ω_s drives the g_2 leg in H_0 ; for branch $+1$ the assignment swaps. These are *physically distinct mode pairs* and must be computed separately — never summed into one susceptibility matrix (see Sec. 7.3 and the README “Traps”).

4.2 The dissipator

The dissipator is a sum of GKSL terms, one per irreversible channel,

$$\mathcal{D}[\rho] = \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} L_k^\dagger L_k \rho - \frac{1}{2} \rho L_k^\dagger L_k \right). \quad (7)$$

Two kinds of channel appear (`atoms.AtomModel._build_lindblad`):

- **Spontaneous emission:** jump operators $L_{e \rightarrow g} = \sqrt{\Gamma_{e \rightarrow g}} |g\rangle\langle e|$ with the CG-branched rates of Eq. (2). The “recycling” term $L \rho L^\dagger$ refeeds the emitted population into the ground states; the anticommutator term damps the optical coherences at $\Gamma/2$ (the homogeneous optical linewidth).
- **Ground-state (Raman) dephasing:** a decay of the $g_1 \leftrightarrow g_2$ coherence, $\dot{\rho}_{g_1 g_2} \supset -\gamma_{gg} \rho_{g_1 g_2}$. This is the rate that sets the width of the Raman/two-photon resonance and ultimately limits the squeezing bandwidth. It is no longer a single constant: it is a transit-time / residual floor ($\gamma_{\text{floor}}/2\pi = 100$ kHz, `constants.GAMMA_GG`) plus a density-dependent Rb–Rb collision term (Sec. 4.3).

- **Optical collisional dephasing:** an extra decay γ_{opt} on every optical (g, e) coherence, representing Rb–Rb self-broadening of the optical line on top of the natural $\Gamma/2$ (Sec. 4.3). With $\gamma_{\text{opt}} = 0$ (the registry default) the natural-linewidth model is recovered — this is what the pinned kernel regressions use.

Remark 1. *The general `AtomModel` also supports polarization-grouped jump operators (Σ_q form) that transfer coherence during spontaneous emission (used by the magneto-optics schemes). The four-level FWM model only needs the population-refeeding form above.*

4.3 Density-dependent collisional decoherence

Sim *et al.* attribute the degradation of squeezing at high vapour temperature to collision-driven decoherence, and the model now carries that physics explicitly. Two dephasing channels grow with density; both are built per temperature by `fwm.collisional_atom(T)`, which calls `atoms.double_lambda_rb85(gamma_opt)`:

1. **Optical self-broadening.** Resonant Rb–Rb collisions broaden the optical line linearly in density,

$$\Gamma_{\text{eff}} = \Gamma + \beta N, \quad \beta/2\pi = 0.69 \times 10^{-7} \text{ Hz cm}^3, \quad (8)$$

(`hyperfine.self_broadened_gamma`; the β value is the one the AutoOD-validated absorption path uses). The natural Γ is already carried by the spontaneous-emission jump operators, so the dissipator receives only the *added* collisional half-width

$$\gamma_{\text{opt}} = \frac{1}{2}(\Gamma_{\text{eff}} - \Gamma) = \frac{1}{2} \beta N \quad (9)$$

as a pure-dephasing rate on each optical (g, e) coherence. At the reference 121 °C ($N = 2.15 \times 10^{19} \text{ m}^{-3}$) this is $\gamma_{\text{opt}}/2\pi \approx 0.74 \text{ MHz}$ — already a $\sim 26\%$ addition to the natural half-width $\Gamma/2$, and it grows exponentially with T through $N(T)$.

2. **Ground Raman collisions.** The ground-coherence relaxation is

$$\gamma_{gg}(T, N) = \gamma_{\text{floor}} + N \sigma \bar{v}_{\text{rel}}, \quad \bar{v}_{\text{rel}} = \sqrt{\frac{8k_B T}{\pi (m/2)}}, \quad (10)$$

(`constants.ground_coherence_dephasing`), with the transit/residual floor $\gamma_{\text{floor}}/2\pi = 100 \text{ kHz}$ anchored at the validated operating point and the Rb–Rb spin-exchange cross-section $\sigma = 1.9 \times 10^{-14} \text{ cm}^2$. At 121 °C the collisional term adds only $\sim 3 \text{ kHz}$ — the transit floor dominates over the whole pure-cell range — but the rate now scales correctly with density.

The physical consequence is a genuine interior optimum in temperature: raising T raises N and hence the parametric gain exponentially, but the same N degrades the optical and Raman coherences that the gain is built from, so the achievable squeezing peaks near 121–131 °C and then *worsens* at higher temperature — reproducing the experimental behaviour instead of tracking the density monotonically. The numerical consequence is that $\bar{\chi}$ is no longer temperature-independent: any scanner that caches the $\bar{\chi}(\delta, \Delta_{\text{eff}})$ table must rebuild it per temperature with this atom (Sec. 5).

4.4 Liouville space (vectorization)

Equation (5) is linear in ρ , so stacking the $n \times n$ matrix ρ into a length- n^2 vector turns the whole right-hand side into a single matrix — the Liouvillian $\mathcal{L}$. The code uses row-major vectorization, $\text{vec}(\rho)_{ni+j} = \rho_{ij}$, for which the commutator becomes (`core.comm_super`):

$$\text{vec}([H, \rho]) = (H \otimes \mathbb{K} - \mathbb{K} \otimes H^T) \text{vec}(\rho), \quad \mathcal{C}[H] \equiv -i(H \otimes \mathbb{K} - \mathbb{K} \otimes H^T). \quad (11)$$

The dissipator (7) vectorizes likewise; each term $L\rho L^\dagger$ becomes $L \otimes L^*$, and the anticommutators become $\frac{1}{2}(L^\dagger L \otimes \mathbb{K} + \mathbb{K} \otimes (L^\dagger L)^T)$ (`atoms._build_lindblad`). The full base Liouvillian is

$$\mathcal{L}_0 = \mathcal{C}[H_0] + \mathcal{D}, \quad (12)$$

(`core.build_liouvillian`). Finding the steady state is then the linear algebra problem $\mathcal{L}_0 \text{vec}(\rho) = 0$ subject to $\text{Tr} \rho = 1$; the trace condition replaces one redundant row of $\mathcal{L}_0$ (row 0, the $\rho_{g_1 g_1}$ equation) by $\sum_i \rho_{ii} = 1$. This “replace a row with the trace” construction recurs in every solver in `core.py`.

5 Thermal motion: the Doppler average

5.1 Velocity classes

At $T \sim 120^\circ\text{C}$ the atoms move fast enough that the Doppler shift kv ($\sigma_v = \sqrt{k_B T/m} \sim 170 \text{ m/s}$ gives $k\sigma_v/2\pi \sim 200 \text{ MHz}$) dwarfs the natural linewidth. The vapour is therefore a Maxwell ensemble of velocity classes (`doppler.velocity_grid`):

$$v \in [-v_{\max}, v_{\max}], \quad v_{\max} = [c_\sigma \sigma_v], \quad w(v) \propto \exp\left(-\frac{v^2}{2\sigma_v^2}\right), \quad \sum_v w(v) = 1, \quad (13)$$

with cutoff $c_\sigma = 3$ standard deviations and a step dv of a few m/s. Each class sees both lasers Doppler-shifted by kv (the pump and the near-degenerate seed/conjugate share essentially the same k , since they differ only by the GHz hyperfine scale $\ll$ the optical frequency).

5.2 The Δ_{eff} trick

The crucial observation (`doppler.py` header, README “Speed”) is that the *only* place the one-photon detuning Δ and the velocity v enter the Hamiltonian is through the excited-state diagonal. Define the effective one-photon detuning

$$\Delta_{\text{eff}} = \Delta - kv. \quad (14)$$

Then the velocity-dependent Liouvillian is a *rigid shift* of one template,

$$\mathcal{L}_0(\Delta_{\text{eff}}) = \mathcal{L}_0(0) - \Delta_{\text{eff}} S_v, \quad S_v = \mathcal{C}\left[\sum_{e \in \text{excited}} |e\rangle\langle e|\right], \quad (15)$$

where S_v is the commutator super-operator of the projector onto the excited manifold (`atoms._build_velocity...`). Consequently:

- In the *Hamiltonian*, temperature enters only through the Maxwell weights $w(v)$, and the detuning Δ only as an offset of the Δ_{eff} axis.
- For a *fixed dissipator* the susceptibility table $\bar{\chi}(\delta, \Delta_{\text{eff}})$ is therefore a function of $(\delta, \Delta_{\text{eff}})$ alone, with no explicit T or absolute- Δ dependence.
- The collisional decoherence of Sec. 4.3 puts $N(T)$ into the dissipator, so the table is now T -dependent through $\gamma_{\text{opt}}(N)$ and $\gamma_{gg}(T, N)$: *one table per temperature*, still reusable across all Δ and all velocity classes at that temperature.

The Δ_{eff} sampling axis is built (`doppler.build_Delta_eff_axis`) to span $\Delta - kv$ across a spectrum call’s velocity grid, so the later average is a cheap index interpolation rather than a re-solve. The dedicated (Δ, T) squeezing map (`analysis/scan_squeezing_delta.T.py`) builds, per temperature, a master Δ_{eff} axis from the exact union of all velocity samples needed by every Δ in the scan and evaluates $\bar{\chi}(\delta, \Delta_{\text{eff}})$ once on that master grid with that temperature’s collisional atom; each map point then only gathers the relevant columns and applies the Maxwell weights.

6 The multimode steady state (truncated Floquet)

6.1 Why the steady state is not static

In ordinary absorption one weak field sits inside H_0 and the steady state is a single static ρ . FWM is different: the seed and the conjugate are *different frequencies*, separated by twice the ground hyperfine splitting, and they couple through the ground-state coherence $\rho_{g_1 g_2}$, which therefore *oscillates* at the beat frequency

$$\Omega_{\text{beat}} = \Omega_{\text{HF}} - \text{branch} \cdot \delta, \quad \Omega_{\text{HF}} = 2\pi\nu_{\text{HF}}. \quad (16)$$

The Hamiltonian is thus time periodic. We split it into the static part H_0 (pump + one weak field, Eq. (6)) and a *sideband* part that oscillates at Ω_{beat} (**sideband_hamiltonian**):

$$H(t) = H_0 + H_p e^{-i\Omega_{\text{beat}}t} + H_p^\dagger e^{+i\Omega_{\text{beat}}t}. \quad (17)$$

H_p contains the *other* weak field plus the pump on the complementary leg; its one-sided form ($|e\rangle\langle g|$ only, **_add_sideband_drive**) is the positive-frequency component of that drive.

6.2 Three-mode ansatz and truncation

Because the drive is weak, the response is dominated by the carrier and the first sidebands. Expand

$$\rho(t) = \rho_0 + \rho_{+1} e^{-i\Omega_{\text{beat}}t} + \rho_{-1} e^{+i\Omega_{\text{beat}}t} + \dots \quad (18)$$

and keep $|n| \leq 1$ (a three-mode truncated Floquet expansion). Substituting into the vectorized master equation and matching harmonics gives, with $C_p \equiv \mathcal{C}[H_p]$ and $C_m \equiv \mathcal{C}[H_p^\dagger]$ (**sideband_template**):

$$(\mathcal{L}_0 + i\Omega_{\text{beat}}) \rho_{+1} = -C_p \rho_0, \quad (19)$$

$$(\mathcal{L}_0 - i\Omega_{\text{beat}}) \rho_{-1} = -C_m \rho_0, \quad (20)$$

$$\mathcal{L}_0 \rho_0 = -(C_p \rho_{-1} + C_m \rho_{+1}). \quad (21)$$

Equation (21) is the carrier steady state *dressed* by the sidebands feeding back into it.

6.3 Feedback elimination

Solving Eqs. (19)–(20) for the sidebands and substituting into (21) eliminates $\rho_{\pm 1}$, leaving a single effective equation for the carrier (**core_floquet_solve**):

$$\boxed{\left[\mathcal{L}_0 - C_p (\mathcal{L}_0 - i\Omega_{\text{beat}})^{-1} C_m - C_m (\mathcal{L}_0 + i\Omega_{\text{beat}})^{-1} C_p \right] \rho_0 = 0.} \quad (22)$$

The two correction terms are exactly the code's **minus_feedback** and **plus_feedback**; the bracket is **A_eff**. Trace-normalizing (row $0 \rightarrow \sum_i \rho_{0,ii} = 1$) and solving gives ρ_0 ; the upper sideband then follows directly,

$$\rho_{+1} = -(\mathcal{L}_0 + i\Omega_{\text{beat}})^{-1} C_p \rho_0, \quad (23)$$

reusing the factor already computed for (22) rather than solving a second system. Physically, ρ_0 holds the optical coherence at the seed frequency (the probe response) and ρ_{+1} holds the coherence at the conjugate frequency (the generated conjugate) — the two outputs we need.

Compiled grid solve. Equation (15), together with the affine δ dependence of the ground-state Hamiltonian, lets the FWM grid be assembled from constant matrices:

$$\mathcal{L}_0(\delta, \Delta_{\text{eff}}) = \mathcal{L}_0(0, 0) + \delta C_\delta - \Delta_{\text{eff}} S_v, \quad \Omega_{\text{beat}}(\delta) = \Omega_{\text{HF}} - \text{branch} \cdot \delta. \quad (24)$$

When Numba is available, `kernels.floquet_chi_grid` evaluates the full $(\delta, \Delta_{\text{eff}})$ grid in a fused parallel loop: assemble the local Liouvillian, factor the two sideband matrices, form A_{eff} , solve the trace-normalized carrier, recover ρ_{+1} , and contract directly with the CG-weighted readout vectors. The NumPy path follows the same equations with a per- δ loop over `core.floquet_solve`; `tests/test_kernels.py` pins the two paths to the same numerical result.

7 The susceptibility matrix

7.1 What $\bar{\chi}$ means

We want the linear response of the two optical polarizations (one per Raman leg) to the two weak fields. Define the dimensionless susceptibility matrix by

$$\begin{pmatrix} P_s \\ P_c \end{pmatrix} = \begin{pmatrix} \bar{\chi}_{ss} & \bar{\chi}_{sc} \\ \bar{\chi}_{cs} & \bar{\chi}_{cc} \end{pmatrix} \begin{pmatrix} \Omega_s \\ \Omega_c \end{pmatrix}, \quad (25)$$

where P_s (P_c) is the CG-weighted optical coherence read out on the seed (conjugate) leg, and Ω_s (Ω_c) is the seed (conjugate) Rabi frequency. The polarization readout is (`_polarization_coherence`):

$$P_{\text{leg } g} = \sum_{e \in \text{excited}} \sqrt{3C_F^2(F_g, F_e)} \rho_{eg}, \quad (26)$$

the same CG weighting as the drive, so absorption and emission are described by one consistent dipole. Because the response is linear in the *weak* fields (the pump is the strong field, already inside H_0 and the sidebands), $\bar{\chi}$ is *independent* of the magnitude of the reference field that probes it — the code simply divides the measured coherence by the reference Rabi Ω_{ref} .

7.2 Two solves per detuning

Each entry of the 2×2 matrix is a response of one leg to one field, so two independent Floquet solves at each two-photon detuning δ extract all four (`chi_matrix_table`):

- **Solve 1 — seed drive only** ($\Omega_c = 0$ in the sidebands, $\Omega_s = \Omega_{\text{ref}}$ in H_0). The carrier coherence on the seed leg gives $\bar{\chi}_{ss} = P_s/\Omega_{\text{ref}}$; the upper-sideband coherence on the conjugate leg gives $\bar{\chi}_{cs} = P_c/\Omega_{\text{ref}}$ — this is the seed $\rightarrow$ conjugate *generation*.
- **Solve 2 — conjugate seed only** ($\Omega_c = \Omega_{\text{ref}}$ in the sidebands, $\Omega_s = 0$). Symmetrically yields $\bar{\chi}_{sc}$ and $\bar{\chi}_{cc}$.

The off-diagonal $\bar{\chi}_{cs}, \bar{\chi}_{sc}$ are the parametric coupling that does not exist without the pump; they are the heart of the four-wave mixing. `chi_matrix_table` accepts the atom model as an argument (`atom=`): `compute_spectrum` passes the per-temperature collisional atom of Sec. 4.3, while the default natural-linewidth `ATOM` keeps the kernel/NumPy regression callers unchanged. In the compiled path, the two physical solves correspond to two calls to `kernels.floquet_chi_grid`: one with the seed reference field and one with the conjugate reference field. Each call fills the whole $(\delta, \Delta_{\text{eff}})$ grid and returns the probe-leg and conjugate-leg coherences already contracted with the proper CG readout weights.

7.3 Detuning conventions and the two Raman branches

The probe is scanned in laboratory frequency, referenced to the $F = 2 \rightarrow F' = 3$ line. The two Raman resonances sit at (`branch_center_GHz`)

$$f_{\text{Raman}}^{(\pm)} = \Delta_{\text{GHz}} \pm \nu_{\text{HF}}, \quad (27)$$

and the two-photon detuning of a probe point is (`two_photon_detuning_from_probe_scan`)

$$\delta = 2\pi(f_{\text{probe}} - f_{\text{Raman}}^{(\text{branch})}). \quad (28)$$

The beat frequency (16) that drives the Floquet expansion uses this same δ . **The two branches are separate physical channels** (separate probe/conjugate mode pairs): the standard squeezing operating point is the red-detuned seed on the $(-)$ branch. Adding the two branches' $\bar{\chi}$ matrices into one propagation is wrong — it manufactures spurious high-gain features (the README documents an artefact near +70 MHz TPD). The wide “both branches” view computes branch -1 and $+1$ independently and overlays them (`full_spectrum`).

8 Doppler averaging of the susceptibility

The single-velocity table $\bar{\chi}(\delta, \Delta_{\text{eff}})$ is averaged over the Maxwell distribution to give the macroscopic susceptibility seen by the beam. Using $\Delta_{\text{eff}} = \Delta - kv$ and the uniform Δ_{eff} axis, the average is a weighted linear interpolation (`doppler.doppler_average`):

$$\bar{\bar{\chi}}(\delta; \Delta, T) = \sum_v w(v) \bar{\chi}(\delta, \Delta - kv). \quad (29)$$

This is a *coherent average of the complex susceptibility*: amplitude and phase from each velocity class add before the field propagates, which is the correct order of operations for a dispersive parametric medium (the alternative — averaging intensities — would discard the inhomogeneous phase that sets the gain). The interpolation indices and fractions are precomputed once (`interpolation_weights`) and applied to every row of the table (`apply_doppler_average`), so the average costs one matrix multiply. In the (Δ, T) squeezing-map script, the per-temperature master Δ_{eff} axis is the exact union of all requested $\Delta - kv$ samples at that temperature. The Doppler step therefore becomes a direct column gather followed by the same weighted sum, with no interpolation error and no additional Floquet solve.

9 Maxwell–Bloch propagation: susceptibility $\rightarrow$ gain

9.1 Coupled amplitude equations and phase conjugation

In a dilute medium the slowly varying seed and conjugate Rabi amplitudes obey coupled propagation equations along the cell (z direction). The combination that closes is $(\Omega_s, \Omega_c^*)^T$ — note the *complex conjugate* of the conjugate field, the signature of a phase-conjugating ($\chi^{(3)}$) process:

$$\frac{d}{dz} \begin{pmatrix} \Omega_s \\ \Omega_c^* \end{pmatrix} = M \begin{pmatrix} \Omega_s \\ \Omega_c^* \end{pmatrix}, \quad M = \frac{i}{2} \begin{pmatrix} k_s \chi_{ss} & k_s \chi_{sc} \\ -k_c \chi_{cs}^* & -k_c \chi_{cc}^* \end{pmatrix}. \quad (30)$$

The diagonal terms are self-action (gain/absorption and dispersion on each beam); the off-diagonal terms are the parametric coupling that converts seed into conjugate and back. The conjugation and the minus signs in the second row are what make M generate *amplification* (a non-unitary, energy-non-conserving transfer) rather than mere rotation.

9.2 Seeded longitudinal phase mismatch

The seeded propagation matrix can carry a rotating-frame longitudinal mismatch term:

$$M_{\Delta k} = M + \frac{i\Delta k_z}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (31)$$

For the seeded double- Λ geometry the vacuum mismatch used in Balanced resolution is

$$\Delta k_z^{(0)}(\delta) = 2k_p - (k_s(\delta) + k_c(\delta)) \cos \theta, \quad \theta = 0.32^\circ, \quad (32)$$

where the reference angle is the Sim *et al.* pump–probe crossing angle, k_s is set by the scanned seed frequency, and energy conservation sets $k_c = 2k_p - k_s$ in angular-frequency units. This is cheap: it is one vector over the already-computed probe axis, and no OBE/Floquet solve is repeated.

High fidelity adds two post-OBE refinements while keeping the expensive susceptibility table fixed:

1. A conservative refractive correction,

$$k_s \rightarrow n_s k_s, \quad k_c \rightarrow n_c k_c, \quad n_{s,c} \simeq 1 + \frac{1}{2} \text{Re} \chi_{ss,cc},$$

with the correction clipped to the dilute-vapour range so resonant dispersive excursions do not create artificial phase-matching runaways.

2. A segmented propagation profile. The cell is sliced into 16 segments; the diagonal self-action remains fixed, while the off-diagonal parametric coupling is smoothly attenuated by a scalar profile inferred from the median absorptive optical depth. This is a propagation-only approximation, not a new atomic solve.

The `phase_detail` option selects how much of this propagation physics is included. `PHASE_LEGACY` sets $\Delta k_z = 0$ for regression-preserving direct calculations; the app’s Fast and Balanced presets use Eq. (32); High fidelity uses the refractive and segmented refinements above; Ultra uses the dynamic-depletion propagation described next.

9.3 Ultra propagation: dynamic pump budget, overlap, diagnostics

Ultra (`_ultra_segmented_gain`) slices the cell into 64 segments and propagates the two-component amplitude explicitly, segment by segment:

- **Refractive phase matching, one pass.** The refractive indices $n_{s,c}(\chi)$ do not depend on Δk_z , so there is no fixed point to iterate: a single refractive update of Eq. (32) is exact (`ULTRA_PHASE_ITERATIONS = 1`). The reported diagnostic is the vacuum→refractive shift of Δk_z , not the last delta of a no-op loop.
- **Gaussian beam overlap.** The pump–probe crossing angle separates the beams along the cell; the off-diagonal coupling in each segment is weighted by the crossed-Gaussian amplitude overlap $\exp[-z^2 \tan^2 \theta / (w_p^2 + w_s^2)]$, normalized to its maximum (`_gaussian_overlap_profile`).
- **Segment-wise pump budget.** After each segment the added seed power and the generated conjugate power are subtracted from the remaining pump budget, and the parametric (off-diagonal) coupling of the next segment is scaled by $\sqrt{P_{\text{remaining}}/P_{\text{pump}}}$ — an approximate *dynamic* pump depletion along z , on top of the global Manley–Rowe cap. The minimum remaining pump along the scan is reported.

- **In-cell loss is not applied here.** The absorptive segment profile attenuates only the parametric coupling; the in-cell *loss* is charged once, downstream, as the beamsplitter vacuum admixture of Sec. 12. (Applying it here as field attenuation too double-counted the same optical depth in an earlier Ultra version.)
- **Zeeman diagnostic.** The 24-level ^{85}Rb D_1 Zeeman manifold is built and its CG sums are checked against the lumped $3C_F^2$ strengths (consistency $\equiv 1.0$ by construction). It is reported as a diagnostic only — deliberately *not* multiplied into the coupling — and the full 24-level Floquet scan is not run in Ultra v1.

9.4 From dimensionless to physical susceptibility

The dimensionless $\bar{\chi}$ becomes a physical susceptibility through the medium coupling constant (`observables.gain_from_chi`):

$$\chi_{xy} = -\frac{2N |d_{\text{eff}}|^2}{\varepsilon_0 \hbar} \bar{\chi}_{xy}, \quad |d_{\text{eff}}|^2 = \ell_s \varrho_{\text{norm}} |d_{D_1}|^2, \quad (33)$$

where N is the number density and d_{D_1} the D_1 reduced dipole. The sign makes a passive transition absorptive ($\text{Im } \chi > 0$ on resonance). The effective dipole factorizes into a *first-principles* normalization and a *residual* calibration:

- $\varrho_{\text{norm}} = p_F/[2(2I + 1)]$ (`fwm.physical_coupling_norm`) is the thermal ground-population fraction of the probed manifold, $p_F = (2F + 1)/12$, divided by the ground Zeeman degeneracy $2(2I + 1) = 12$ ($I = 5/2$). The AutoOD-validated absorption path ($< 0.1\%$ against measured cell transmission) carries exactly this factor; the lumped 4-level solve instead normalizes $\text{Tr } \rho = 1$ over just four levels and multiplies by the *total* atomic density, so the factor would otherwise be missing from the macroscopic scale. The seeded probe couples from $F = 3$ on the standard ($-$) Raman branch, giving $\varrho_{\text{norm}} = \frac{7}{12} \cdot \frac{1}{12} = \frac{7}{144} \approx 0.0486$ (and $\frac{5}{144} \approx 0.0347$ on the $(+)$ branch). Quantitatively, this is the factor the old hand-tuned fudge “line_strength = 0.05” was standing in for.
- ℓ_s (`LINE_STRENGTH_FACTOR`; the app’s Advanced “line-strength factor”) is the dimensionless *residual* left on top, engine default 1.0. The seeded app default is $\ell_s = 0.74$, anchored so that Ultra fidelity reproduces the measured $G_s \approx 14$ and -7.8 dB at the Sim *et al.* operating point; the departure from 1 stands for the m -sublevel participation / beam-geometry reduction not yet derived from first principles. It is not an experimentally tunable quantity, which is why the app parks it under *Advanced*.

The bookkeeping is now clean: the genuine hyperfine CG structure is inside $\bar{\chi}$ (drive $\times$ readout $= 3C_F^2$ per leg), the population/degeneracy normalization is ϱ_{norm} , and the density is the validated CRC value — the same normalization chain that reproduces measured linear absorption. ℓ_s rescales the *effective* $|d|^2$ in propagation only; it does *not* touch the OBE solve (where Γ and Ω are fixed by I_{sat}).

9.5 The transfer matrix

Because M is (per detuning point) a constant 2×2 matrix over the cell length L , the propagation is solved exactly by the matrix exponential

$$T = \exp(ML), \quad \begin{pmatrix} \Omega_s \\ \Omega_c^* \end{pmatrix}_{z=L} = T \begin{pmatrix} \Omega_s \\ \Omega_c^* \end{pmatrix}_{z=0}. \quad (34)$$

The 2×2 exponential is evaluated in closed form (`core.matrix_exp_2x2`; derivation in Appendix B) for the whole batch of detuning points at once. The power gains are

$$G_s = |T_{00}|^2 \quad (\text{seed transmission}), \quad G_c = |T_{10}|^2 \quad (\text{conjugate generated from a seed at port 0}). \quad (35)$$

With a seed in and vacuum in the conjugate port, G_s is the amplified seed power and G_c is the newly generated conjugate power. In the high-gain parametric regime the twin-beam relation $G_c \simeq G_s - 1$ holds (each created conjugate photon accompanies one added seed photon).

The exponential-density trap. Since $M \propto N$ (Eq. (33)) and $T = e^{ML}$, the gain grows *exponentially* with density, i.e. super-exponentially with temperature through $N(T)$. This is physically correct only while the pump is undepleted; pushed too far it over-amplifies, which the next section caps.

Overflow clamp. Numerically, `core.matrix_exp_2x2` clips the *real part* of the exponent arguments cL and sL at ± 350 — two such factors multiply, and float64 e^x overflows near $x \simeq 709$. Without the clamp, a runaway linear gain returns `inf`, and the downstream pump-depletion saturation then evaluates $\infty/(1 + \infty) = \text{NaN}$, silently poisoning the whole gain/squeezing curve. With it, an unphysical linear gain stays huge-but-finite and is capped to the energy bound by the saturation. The clamp is a strict no-op in the validated regime ($|\text{Re}(cL)| \ll 350$).

10 Pump depletion: the Manley–Rowe cap

The propagation above is linear in the *undepleted* pump, so at high density it can predict $G_s P_{\text{seed}} \gg P_{\text{pump}}$ — more output power than the pump can supply. Non-degenerate FWM is a Manley–Rowe process: two pump photons make one signal photon and one conjugate photon, so at full conversion the added signal power and the generated conjugate power each approach half the pump power. The code enforces this energy bound with a smooth homogeneous-saturation form (`observables.pump_depletion_saturation`):

$$G_s^{\text{sat}} = 1 + \frac{G_s - 1}{1 + (G_s - 1) P_{\text{seed}}/P_{\text{cap}}}, \quad G_c^{\text{sat}} = \frac{G_c}{1 + G_c P_{\text{seed}}/P_{\text{cap}}}, \quad P_{\text{cap}} = \frac{P_{\text{pump}}}{2}. \quad (36)$$

Three properties make this the right minimal fix:

- **Small-signal limit.** When $(G - 1)P_{\text{seed}} \ll P_{\text{cap}}$ the denominators are ≈ 1 and the validated gain is untouched.
- **Energy ceiling.** As $G \rightarrow \infty$, $(G_s^{\text{sat}} - 1)P_{\text{seed}} \rightarrow P_{\text{cap}}$ and $G_c^{\text{sat}}P_{\text{seed}} \rightarrow P_{\text{cap}}$ — exactly the half-pump bound.
- **Twin-beam relation preserved.** $(G_s - 1)$ and G_c saturate identically, so $G_c^{\text{sat}} \simeq G_s^{\text{sat}} - 1$ still holds and the squeezing formula stays consistent.

The pipeline keeps the pre-saturation small-signal peak for diagnostics and flags the operating point as “pump-depletion limited” when the small-signal gain exceeds the cap by more than 10 % (`compute_spectrum, _seeded_observables`).

11 Quantum observables of the twin beam

11.1 Intensity-difference squeezing

The twin beams are correlated, so the noise in $\hat{n}_s - \hat{n}_c$ can fall below the shot-noise level. For the lossless seeded parametric amplifier the correct starting point is the two-mode Bogoliubov transformation $\hat{a}_s \rightarrow u \hat{a}_s + v \hat{a}_c^\dagger$, $\hat{a}_c \rightarrow u \hat{a}_c + v \hat{a}_s^\dagger$ with $|u|^2 - |v|^2 = 1$, i.e. $G_s = |u|^2$, $G_c = |v|^2 = G_s - 1$. This transformation *conserves the photon-number difference*: $\hat{n}_s - \hat{n}_c$ commutes with the two-mode squeeze operator, so

$$(\hat{n}_s - \hat{n}_c)_{\text{out}} = (\hat{n}_s - \hat{n}_c)_{\text{in}}. \quad (37)$$

With a bright coherent seed in port s (Poissonian, $\text{Var} = \langle n \rangle_{\text{in}}$) and vacuum in port c , the output difference noise is just the *input* seed shot noise, while the shot-noise reference of the output beams is $\langle \hat{n}_s + \hat{n}_c \rangle_{\text{out}} = (G_s + G_c) \langle n \rangle_{\text{in}}$. Generalizing the prefactor to unequal gains (which the propagation over absorptive/dispersive χ produces), the ideal noise implemented in `observables.ideal_twin_beam_noise` is

$$S_{\text{ideal}} = \frac{(G_s - G_c)^2}{G_s + G_c} \xrightarrow[G_s - G_c = 1]{\text{lossless}} \frac{1}{G_s + G_c} = \frac{1}{2G_s - 1}, \quad (38)$$

the standard result of the FWM squeezing literature (Sim *et al.*; McCormick *et al.*, *Opt. Lett.* **32**, 178 (2007)). At unity gain $S_{\text{ideal}} = 1$ (shot noise); at high gain $S_{\text{ideal}} \rightarrow 1/(2G_s - 1) \rightarrow 0$. The value is clipped to $[0, 1]$ so that passive-loss regions of a probe scan — no parametric gain, $G_s - G_c$ arbitrary — cannot spuriously report sub-shot-noise values.

Remark 2 (The corrected formula). *Until 2026-07-01 the code used $S_{\text{old}} = (G_s + G_c - 2\sqrt{G_s G_c - 1}) / (G_s + G_c)$. The two expressions agree only at $(G_s, G_c) = (2, 1)$. On the twin-beam line $G_c = G_s - 1$ the old numerator is a catastrophic cancellation whose expansion gives $S_{\text{old}} \simeq \frac{5}{2} / (2G_s - 1)^2$ instead of $1/(2G_s - 1)$ — a spurious extra squeezing factor $\approx (2G_s - 1)/2.5$ that grows with gain (at $G_s \sim 15$, about +10 dB of fictitious squeezing). Under normal operation the error was masked because the detection floor $(1 - \eta)$ dominates S , but it corrupted every ideal-detection study: re-evaluating the same (Δ, T) ideal-scan data gives -84.7 dB with the old formula versus -31.3 dB with the corrected one (squeezing report v4). The pinned regression baseline was re-captured with the corrected formula.*

Symmetric detection efficiency η on both arms mixes in uncorrelated vacuum,

$$S(\eta) = \eta S_{\text{ideal}} + (1 - \eta), \quad S_{\text{dB}} = 10 \log_{10} S, \quad (39)$$

with $\eta = \text{QE}(1 - \text{loss})$. Equation (39) immediately gives the squeezing *floor*: since $S_{\text{ideal}} \geq 0$,

$$S_{\text{dB}} \geq 10 \log_{10}(1 - \eta), \quad (40)$$

a hard limit set purely by loss. At the reference detection chain ($\text{QE} = 0.92$, 5.5% optical loss, $\eta = 0.8694$) the floor is -8.84 dB. (The 2-D $\xi(\Delta, T)$ scan confirms the model saturates exactly at this plateau once the gain is large enough — gain saturation and the loss floor together cap the achievable squeezing.) Negative S_{dB} means squeezed below shot noise; the regression-pinned legacy scan at the Sim *et al.* reference setting reaches $S_{\text{dB}} \approx -8.69$ dB at $G_s \approx 32$ (Sec. 13).

11.2 Beyond symmetric loss: the balanced two-arm noise model

Real detection is rarely symmetric: the two arms can have different total transmissions, and the reference photocurrent can be electronically weighted. The generalized noise (`observables.balanced_twin_beam`

is the variance of the weighted difference photocurrent, normalized to the coherent-state shot noise of the same combination:

$$D = I_s - w I_c, \quad S = \frac{\text{Var } D}{\langle I_s \rangle + w^2 \langle I_c \rangle}, \quad \langle I_s \rangle = \eta_s G_s, \quad \langle I_c \rangle = \eta_c G_c, \quad (41)$$

with total arm efficiencies η_s, η_c (cell transmission $\times$ path loss $\times$ QE) and

$$\text{Var } D = \eta_s G_s + w^2 \eta_c G_c - 2w \eta_s \eta_c \text{cov}_0, \quad \text{cov}_0 = \frac{1}{2} \left[(G_s + G_c) - (G_s - G_c)^2 \right], \quad (42)$$

where cov_0 is the source intensity covariance consistent with the conserved-difference result: at $w = 1$, $\eta_s = \eta_c = \eta$ the model reduces *exactly* to Eqs. (38)–(39), and on the lossless twin-beam line $\text{cov}_0 = G_c$. It is clipped to the Cauchy–Schwarz bound $\sqrt{G_s G_c}$ (and to ≥ 0). The reference weight w may be a scalar electronic gain or one of three lab conventions: **raw** ($w = 1$), **dc** ($w = \langle I_s \rangle / \langle I_c \rangle$, cancelling the mean photocurrents — what a balanced detector with a DC servo does), and **shot** ($w = \sqrt{\langle I_s \rangle / \langle I_c \rangle}$, minimizing the normalized quantum noise at fixed detected powers). An optional **source_noise** input replaces the ideal source covariance with a measured/inferred value (used by paper-anchored resonant-FWM readouts elsewhere in GABES), and additive technical excess-noise terms enter as $S \rightarrow S + \mathcal{N}_{\text{tech}}$. This model is the noise engine of the hardened Ultra path (Sec. 12).

11.3 Twin-beam photon statistics

In the same ideal-parametric spirit, the equal-time photon statistics follow from the mean pair number $n = G_s - 1$ (`observables.coincidence_stats`):

$$g_{ss}^{(2)} = g_{cc}^{(2)} = 2 \text{ (each arm thermal)}, \quad g_{sc}^{(2)}(0) = 2 + \frac{1}{n}, \quad R = \frac{[g_{sc}^{(2)}]^2}{g_{ss}^{(2)} g_{cc}^{(2)}} = \frac{(2 + 1/n)^2}{4}. \quad (43)$$

The Cauchy–Schwarz ratio $R > 1$ everywhere there is net gain is the nonclassical photon-pair signature; $g_{sc}^{(2)} \rightarrow 2$ at high flux and diverges at low flux. These are valid only where $G_s > 1$ (set to NaN otherwise) and, like the squeezing above, do not include quantum-Langevin propagation loss.

12 The Ultra noise model: in-cell loss, asymmetric arms, pump scatter

The legacy/Balanced/Fine paths score squeezing with the symmetric formula Eq. (39). The Ultra path models the noise budget more honestly, in two stages: a beamsplitter treatment of in-cell loss (pre-hardening Ultra), and — intrinsic since the v5 hardening — an excess-noise model that charges the penalties a gain-gap-only objective is blind to.

12.1 In-cell loss as a beamsplitter

Propagation loss inside the cell happens *before* the detectors, so Ultra applies it to the source noise as a vacuum admixture with transmission $\tau_{\text{cell}} = e^{-\text{OD}}$ (from the median absorptive optical depth of the segment profile), then the detection efficiency, then any additive technical noise (`observables.segmented_loss_noise_squeezing_dB`):

$$S_{\text{cell}} = \tau_{\text{cell}} S_{\text{ideal}} + (1 - \tau_{\text{cell}}), \quad S = \eta S_{\text{cell}} + (1 - \eta) + \mathcal{N}_{\text{tech}}. \quad (44)$$

Deliberately, this loss is *not* also applied as field attenuation inside the segmented propagation — doing both double-counts the same optical depth (an early Ultra bug). The technical-noise channels ($\mathcal{N}_{\text{tech}}$: seed excess noise, pump scatter, EOM residual) were zero-default placeholders in this stage; Eq. (44) with $\mathcal{N}_{\text{tech}} = 0$ is what `excess_noise_model=False` still computes, kept for A/B against pre-hardening results.

12.2 What a gap-only objective is blind to

$S_{\text{ideal}} = (G_s - G_c)^2 / (G_s + G_c)$ sees only the *gain gap*: any operating point with $G_s - G_c \approx 1$ looks near-perfect regardless of how large the gains are, where the two sidebands sit relative to the absorption manifolds, and how much pump light reaches the detectors. Squeezing report v5 documents the consequence: with probe-only in-cell loss the (Δ, T) optimizer rewarded a near-resonance ultra-high-gain region ($G_s \sim 636$ at an in-cell OD reported as ≈ 0 — an internal contradiction of the lumped model) and pinned a spurious global optimum at $\Delta = -2.20$ GHz, $T = 135^\circ\text{C}$.

12.3 The hardened excess-noise model

The hardened model (intrinsic to Ultra; `compute_spectrum` argument `excess_noise_model`, default on) adds exactly the two missing penalties.

(i) Arm-resolved linear absorption. The lumped double- Λ parametric susceptibilities χ_{ss}, χ_{cc} carry the FWM response but *not* the ordinary linear absorption a real beam suffers when its optical frequency crosses the populated $F = 2 \rightarrow F'$ manifold — the physics that kills squeezing when a sideband lands near resonance (the conjugate in the deep-red- Δ region, the probe in the deep-blue- Δ region; Sim *et al.* document the conjugate case). `fwm.arm_linear_od` evaluates each arm’s own linear optical depth from the AutoOD-validated hyperfine absorption ($< 0.1\%$ vs lab), summing the manifold components with $F_g = 2$ (branch -1) at the arm’s own frequency; the conjugate frequency follows from energy conservation, $f_c = 2\Delta_{\text{GHz}} - f_{\text{probe}}$. These enter as *asymmetric arm transmissions* in the balanced model of Sec. 11.2:

$$\eta_s^{\text{arm}} = \eta e^{-(\text{OD}_{\text{seg}} + \text{OD}_s^{\text{lin}})}, \quad \eta_c^{\text{arm}} = \eta e^{-\text{OD}_c^{\text{lin}}}, \quad (45)$$

where OD_{seg} is the χ_{ss} -based in-cell depth already used by the segment profile and $\text{OD}_{s,c}^{\text{lin}}$ are the manifold linear depths (clipped to $[0, 5]$). Asymmetric arm loss is precisely what breaks the intensity-difference correlation, so parking either sideband on a populated manifold is now charged its physical price.

(ii) Pump-scatter excess noise. The pump’s own linear optical depth OD_p at its detuning Δ (same validated absorption path) fixes how much pump power the vapour scatters; a residual fraction κ of that reaches the difference photocurrent (finite PBS extinction and spatial-filter leakage):

$$\mathcal{N}_{\text{ps}} = \kappa (1 - e^{-\text{OD}_p}), \quad \kappa = 0.1 \text{ (default, the one residual coefficient)}. \quad (46)$$

This is small far from resonance (the $+0.9$ GHz window of Sim *et al.*) and large when the pump sits near a populous manifold (the -2.1 GHz/ $F=3$ region).

Assembled noise. The Ultra squeezing is then the balanced two-arm noise with the DC-cancelling weight and the pump scatter as additive excess noise:

$$S_{\text{Ultra}} = \text{Var}(I_s - wI_c) / (\langle I_s \rangle + w^2 \langle I_c \rangle) \Big|_{\eta_s^{\text{arm}}, \eta_c^{\text{arm}}, w=\text{dc}} + \mathcal{N}_{\text{ps}}, \quad S_{\text{dB}} = 10 \log_{10} S_{\text{Ultra}}. \quad (47)$$

The absorption channels carry *no* tuning coefficient (they are validated physics); κ is the single residual.

12.4 What hardening changes (squeezing report v5)

- **The extreme-gain artefact is removed.** The (Δ, T) global-optimum gain drops $G_s = 636 \rightarrow 27$; the v4 artefact coordinate $(-2.20 \text{ GHz}, 135^\circ\text{C})$ is suppressed from -8.83 to -5.41 dB by the conjugate arm's absorption.
- **The honest optimum stays slightly red-detuned.** The hardened global optimum is $\Delta = -2.0 \text{ GHz}$, $T = 120^\circ\text{C}$, $\xi = -8.07 \text{ dB}$ at $G_s = 27$, with the Sim-side lobe ($\Delta = +1.3 \text{ GHz}$, -7.79 dB) only 0.28 dB shallower — near-degenerate. The residual red preference is real physics *of this model*: the more populated $F = 3$ ground manifold (7/12 vs 5/12) and the spectrally cleaner far-red probe position. What the lumped steady-state model cannot contain is the optical-pumping depletion of $F = 3$ when the pump parks near the $F = 3$ manifold, which is the leading candidate for why the experiment prefers $+0.9 \text{ GHz}$.
- **Even the ideal-detection limit is capped.** With $\eta \rightarrow 1$ the gap-only objective runs away to -36.6 dB at the $G_s = 636$ coordinate; the hardened model caps the ideal optimum at -15.5 dB / $G_s = 27$.

The legacy/Balanced/Fine squeezing formulas and the pinned regression baseline are untouched by the hardening; `excess_noise_model=False` reproduces pre-hardening Ultra, and a dict (e.g. `{"pump_scatter_kappa": 0.2}`) tunes κ .

13 The seeded pipeline, end to end

Collecting the pieces, `compute_spectrum` (one one-photon detuning $\Delta = 2\pi D_{\text{GHz}} \cdot 10^9$) runs:

1. Convert powers to Rabi frequencies, Eq. (3); CRC density $N(T)$, Eq. (4); coupling scale $\ell_s \varrho_{\text{norm}}$ (Sec. 9.4).
2. Build the per-temperature collisional atom (`collisional_atom`: $\gamma_{gg}(T, N)$, $\gamma_{\text{opt}}(N)$, Sec. 4.3).
3. Build the probe-frequency scan axis around the chosen Raman branch, and map each point to a two-photon detuning δ (Sec. 7.3).
4. Build the velocity grid and the Δ_{eff} axis (Sec. 5).
5. Build the $\bar{\chi}$ table on $(\delta, \Delta_{\text{eff}})$ with that atom (Secs. 6–7); with Numba available this is the fused `kernels.floquet_chi_grid` path, otherwise the NumPy `core.floquet_solve` reference path.
6. Doppler-average each of the four $\bar{\chi}$ entries, Eq. (29).
7. Maxwell–Bloch propagate $\rightarrow G_s, G_c$, Eqs. (30)–(35). From Balanced fidelity up the propagation matrix carries the seeded phase-mismatch term, Eq. (31); High fidelity adds the refractive correction and the 16-segment absorption profile; Ultra runs the 64-segment dynamic depletion path with Gaussian overlap (Sec. 9.3).
8. Apply the Manley–Rowe cap, Eq. (36).
9. Squeezing $S_{\text{dB}}(\delta)$: Eq. (39) on the legacy/Balanced/Fine paths; the hardened balanced-arm noise Eq. (47) on Ultra.

The returned dictionary carries G_s , G_c , S_{dB} versus probe detuning, plus derived diagnostics: N , σ_v , Ω_{pump} , Ω_{seed} , the small-signal peak and the depletion cap, the first-principles `coupling_norm` ϱ_{norm} , the `line_strength_residual` ℓ_s and their product (`effective_line_strength`), the active phase-detail mode, Δk_z (operating and vacuum), segment/overlap/pump-budget diagnostics, the Zeeman consistency status, and — on Ultra — the `hardened_noise` block (κ , pump OD, per-arm linear ODs). The TPD slider in the app reads a single operating point off this curve by interpolation (`operating_point`) — instant, no re-solve — which is why the δ knob recomputes nothing.

Anchor. The direct `compute_spectrum` path remains regression-anchored (`tests/test_regression.py`, baseline `tests/baseline_focused.npz`; re-captured after the coupling normalization, the collisional dissipator, and the IDS-formula correction). At the Sim *et al.* reference setting ($\Delta = 0.9$ GHz, $T = 121$ °C, 600 mW pump, $8\ \mu\text{W}$ seed, $\ell_s = 0.74$, 5.5 % loss, $\eta = 0.8694$), the pinned legacy focused scan reaches $S_{\text{dB}} \approx -8.69$ dB at $G_s \approx 32.1$, against the -8.84 dB detection floor of Eq. (40). The same residual $\ell_s = 0.74$ makes the *Ultra* path reproduce the paper’s measured operating point ($G_s \approx 14$, -7.8 dB) — that is the calibration statement; the legacy path, which omits the propagation refinements, runs hotter. Regression tests hold the seeded gain/squeezing arrays to the frozen baseline bit-for-bit, and the app presets select the phase-detail mode per fidelity (Fast/Balanced $\rightarrow$ `balanced`, High $\rightarrow$ `fine`, Ultra $\rightarrow$ `ultra`).

14 Spontaneous FWM: the biphoton mode

The second mode (`GenericFWMSolver`) drops the seed entirely and estimates the *spontaneous* biphoton source: pairs generated from vacuum in a cascade or diamond level topology. It now carries two source models (`biphoton_model` toggle): the default **Predictive** path solves the Doppler-averaged biphoton amplitude from first principles — waveform shape, temporal width, source bandwidth and their parameter dependences are *computed* — while **Calibrated** is the legacy reference-injected estimate kept for comparison. In both, the absolute pair rate stays reference-anchored with physical scaling: the lumped model does not pin the absolute collection coefficient, for the same reason the squeezing mode keeps the ℓ_s residual (Sec. 9.4).

14.1 Topologies and fields

A topology is a set of `LevelSpec` levels and `FieldSpec` fields (pump, coupling, signal, idler), each with a wavelength, propagation direction, and phase sign. The driven-field Rabi frequencies are scaled from per-topology reference anchors as $\sqrt{P/P_{\text{ref}}}$ (`beam.anchored_rabi_mhz`), so the power knobs move the physics without pretending to an absolute beam-geometry calibration. Three topologies are provided:

- **^{87}Rb telecom cascade** $5S_{1/2} \rightarrow 5P_{3/2} \rightarrow 4D_{5/2}$ (780/1529 nm), calibrated to $\sim 38\,000$ cps/mW, $g^{(2)} \approx 44$, OD ≈ 112 , bandwidth ≈ 300 MHz.
- **^{133}Cs “BTW” cascade**, two channels (852–917 nm and 852–795 nm) that differ mainly in their temporal width.
- **Generic diamond** four-level template with user wavelengths (no validated anchor).

14.2 Energy and vector momentum conservation

Energy conservation across the four fields is a bookkeeping check (`energy_mismatch_hz`):

$$\Delta\nu = \nu_{\text{pump}} + \nu_{\text{coupling}} - \nu_{\text{signal}} - \nu_{\text{idler}} \approx 0. \quad (48)$$

Momentum conservation (phase matching) is evaluated as a vector mismatch (`phase_mismatch_vector`). The longitudinal component remains calibrated to the reference geometry,

$$\Delta k_z^{\text{rel}} = \Delta k_z^{\text{abs}} - \Delta k_{\text{ref}}, \quad \Delta k_z^{\text{abs}} = \sum_{\text{fields}} s_i k_i \cos \theta_i, \quad k_i = \frac{2\pi}{\lambda_i}, \quad (49)$$

with phase signs s_i and collection angles θ_i . The reference Δk_{ref} is applied only to the longitudinal component. The transverse component is absolute:

$$\Delta k_x = \sum_{\text{fields}} s_i k_i \sin \theta_i \sigma_i, \quad \sigma_i \in \{-1, 0, +1\}. \quad (50)$$

Pump and coupling fields use $\sigma_i = 0$ in the current biphoton collection model. Signal and idler use the selected collection side. Thus a longitudinally calibrated but transversely inconsistent setting, such as forcing the ^{87}Rb telecom source to $1.5^\circ/1.5^\circ$ on the same side, carries a large transverse mismatch and is suppressed by the sinc weight. The finite interaction length L turns this into a collection efficiency through the standard phase-matching sinc (`phase_matching_weight`):

$$\Delta k_{\text{vec}} = \sqrt{(\Delta k_z^{\text{rel}})^2 + \Delta k_x^2}, \quad \Phi_{\text{vec}} = \text{sinc}^2\left(\frac{\Delta k_{\text{vec}} L}{2}\right) = \left[\frac{\sin(\Delta k_{\text{vec}} L/2)}{\Delta k_{\text{vec}} L/2}\right]^2. \quad (51)$$

Scanning the idler angle traces out the one-dimensional acceptance lobe $\Phi_{\text{vec}}(\theta_{\text{idler}})$. In Fine phase-detail mode, the same vectorized geometry is evaluated on a signal–idler grid (`phase_mismatch_vector_grid`) producing a two-dimensional acceptance map $\Phi_{\text{vec}}(\theta_s, \theta_i)$ without touching the velocity-class source calculation.

14.3 Predictive first-principles waveform

The predictive model (`_predictive_waveform`) evaluates the biphoton temporal amplitude as a *coherent sum over Maxwell velocity classes* — the hot-vapour analogue of the joint spectral amplitude — following Kim *et al.* [*Quantum Sci. Technol.* **9**, 045006 (2024), Eq. (2)], Chen *et al.* [*Phys. Rev. Research* **4**, 023132 (2022), Eqs. (3–5)] and Du, Wen, Rubin [*J. Opt. Soc. Am. B* **25**, C98 (2008)]. Each velocity class contributes the two-photon amplitude

$$\kappa(v) = w(v) \frac{\Omega_p \Omega_c}{4f_1(v)f_2(v) + \Omega_c^2} \propto \text{OD}, \quad \begin{aligned} f_1(v) &= \frac{\Gamma_i}{2} + i(\Delta_p - k_p v), \\ f_2(v) &= \frac{\Gamma_e}{2} + i(\Delta_{2\gamma} - k_{\text{res}} v), \end{aligned} \quad (52)$$

with Γ_i, Γ_e the intermediate/upper natural linewidths, $\Delta_{2\gamma} = \Delta_p + \Delta_c$ the two-photon detuning, and $k_{\text{res}} = k_p + k_c$ (signed by propagation directions) the residual two-photon Doppler wavevector. Three points carry the physics:

- **The Ω_c^2 Autler–Townes term** in the denominator is kept — no weak-coupling approximation — so a strong coupling laser splits the two-photon resonance and reshapes (shortens) the waveform, as in the references.
- **The nonlinear response is $\propto \text{OD}$:** $\kappa \propto \alpha$, so the pair *rate* scales as OD^2 before collection factors.
- **Two-photon (ground) dephasing** $\gamma_g = 0.02 \Gamma_i$ (`GROUND_DEPHASING_FRAC`; Chen *et al.* fit $\gamma \approx 0.02\text{--}0.03 \Gamma$) enters the longitudinal response below.

The time-domain nonlinear response is the coherent velocity sum with the single-photon Doppler phase and the *natural* decay envelope of the idler-emitting intermediate state — no injected lifetime:

$$\tilde{\kappa}(\tau) = \left[\sum_v \kappa(v) e^{ik_p v \tau} \right] e^{-\Gamma_i \tau/2}, \quad \psi(\tau) = \tilde{\kappa}(\tau) \sqrt{\Phi_{\text{vec}}}, \quad (53)$$

where the velocity sum is a Doppler-dephasing integral (classes accumulate phase $k_p v \tau$ and wash out) whose interference sets the beat-then-decay biphoton temporal-waveform (BTW) shape; the wavelength-dependent width *ordering* of the two Cs BTW channels (917 narrower than 795) emerges from this sum rather than being injected. The source bandwidth is measured from the waveform’s own spectrum, and a regime diagnostic compares the Du group-delay time $\tau_g = (2\gamma_g/\Omega_c^2) \text{OD} \Gamma_e$ against the Rabi time $2\pi/\Omega_c$ (“group-delay” vs “damped-Rabi” regime). Chen’s ultimate spectral-brightness ceiling, $\approx (\pi/2) \times 10^6 \text{ pairs s}^{-1} \text{ MHz}^{-1}$, is reported alongside.

Optional OD reshaping (off by default). At high optical depth the pair propagates through its own EIT/slow-light medium: in the conjugate (frequency) domain the joint amplitude is multiplied by the longitudinal function $\Phi(\delta) = \text{sinc } \varrho e^{i\varrho}$ [Du Eq. (14)], with $\varrho(\delta)$ the OD-weighted complex EIT phase [Chen Eq. (5)],

$$\varrho(\delta) = \frac{\text{OD} \Gamma_e}{2} \sum_v w(v) \frac{\delta + i\gamma_g}{\Omega_c^2 - 4(\delta + i\gamma_g)(\delta + \Delta_{2\gamma} - k_{\text{res}}v + i\Gamma_e/2)}, \quad (54)$$

implemented as an FFT convolution of $\tilde{\kappa}(\tau)$ (`_apply_longitudinal_response`), with ϱ clipped to the dilute-regime range. At $\text{OD} \rightarrow 0$, $\Phi \rightarrow 1$ and $\psi \rightarrow \tilde{\kappa}$ exactly. The path is a diagnostic opt-in (`biphoton_od_reshaping`) because the lumped model overestimates the group-delay broadening at high OD and would smear the validated narrow telecom BTW.

14.4 Velocity-grid auto-refinement

The velocity sum in Eq. (53) is a discretized Fourier integral of the *narrow* source resonance $|\kappa(v)|$, whose velocity-space width is of order Γ/k — a few m/s, far narrower than the Doppler width σ_v the navigate-only `biphoton_velocity_step` knob is sized for. A step coarser than that resonance *aliases* the sum: before the fix, the reported telecom BTW width swung by a factor ~ 20 , non-monotonically, as the user step varied over 1–12 m/s — numerics masquerading as physics. The predictive path therefore treats the user step only as an upper bound (`_converged_predictive_waveform`):

1. Measure the actual resonance FWHM of $|\kappa(v)|$ on a fine probe grid (so Autler–Townes and detuning broadening are captured), and start from $dv = \text{FWHM}/16$ (`PRED_V_OVERSAMPLE`).
2. Halve the step until the $|\psi|^2$ FWHM is stable to 3% (`PRED_V_FWHM_TOL`), up to 10 halvings.
3. If the 40 000-point cap is reached first, flag the width as *unconverged* (qualitative only) in the readout.

With this in place the ^{87}Rb telecom predictive width converges to $\approx 0.50 \text{ ns}$ against the measured $0.56(4) \text{ ns}$, and the Cs channels land within $\sim 30\%$ — the remaining gap is the deferred per-source absolute-width calibration (`docs/checklist.json`).

14.5 Absolute rate and optical depth: anchored, physically scaled

The absolute pair rate is reference-anchored with physical scaling (`compute_biphoton`):

$$R_{\text{pair}} = R_{\text{ref}} P_{\text{pump}} [\text{mW}] \sqrt{s_c} \Phi_{\text{vec}} \sqrt{N/N_{\text{ref}}}, \quad (55)$$

with R_{ref} the per-topology anchor (e.g. 38 000 cps/mW for the telecom cascade), s_c the relative coupling drive, Φ_{vec} the vector phase-matching weight of Eq. (51), and $N(T)$ the isotope density. The optical depth that drives Eqs. (52) and (54) is likewise the *measured* reference OD scaled

by density and cell length, $\text{OD} = \text{OD}_{\text{ref}}(N/N_{\text{ref}})(L/L_{\text{ref}})$ — OD is an experimentally reported quantity in these sources, like the cell temperature. Where a topology carries no reference OD, a physical leg OD is computed from the natural linewidth via the AutoOD-validated $\Gamma \rightarrow |d|^2$ route, $\alpha_0 = 2Nk|d|^2/(\varepsilon_0\hbar\Gamma)$ (`species.reduced_dipole_sq`).

14.6 Calibrated (legacy) source model

The Calibrated model keeps the pre-2026-06-19 estimate for comparison: the velocity-class source is the weak-drive product of the two cascade resonances,

$$A(v) = w(v) \frac{\Omega_p \Omega_c}{\left[\frac{\gamma_\ell}{2} + i(\Delta_p - k_p v)\right] \left[\frac{\gamma_u}{2} + i(\Delta_{2\gamma} - k_{\text{res}} v)\right]} \sqrt{\Phi_{\text{vec}}}, \quad (56)$$

summed with the idler-arm phase $e^{ik_{\text{idler}}v\tau}$ and multiplied by an *injected* exponential lifetime envelope $e^{-\tau/\tau_{\text{decay}}}$ (the topology’s reference emission decay); the source bandwidth is the reference value and the $g^{(2)}$ peak is forced to the reference anchor (below). Because the width is set by the injected lifetime rather than the velocity-sum dephasing, this model is insensitive to the velocity-step aliasing of Sec. 14.4.

14.7 Correlation readout and calibration

The detector model (`observables.biphoton_stats`) folds in finite filter transmission $\eta_f = \min(1, \text{BW}_{\text{filter}}/\text{BW}_{\text{source}})$, detector efficiencies $\eta_{s,i}$, dark counts, a coincidence window, and timing jitter (Gaussian smoothing of $|\psi|^2$). It then forms singles, true and accidental coincidences, and the normalized cross-correlation

$$g_{SI}^{(2)}(\tau) = 1 + \text{CAR} \cdot \frac{|\psi(\tau)|^2}{\max |\psi|^2}, \quad \text{CAR} = \frac{\text{coincidence}}{\text{accidental}}. \quad (57)$$

Heralding efficiencies and the Cauchy–Schwarz ratio $R = [g_{\text{peak}}^{(2)}]^2/4$ follow. The two source models differ in one honest detail: in the *Calibrated* model, where a paper anchor exists (e.g. the ^{87}Rb $g^{(2)} \approx 44$), an explicit added-accidental term calibrates the peak to the reference value — the absolute noise floor is *calibrated*, not derived. The *Predictive* model passes no $g^{(2)}$ target: its peak follows from the computed waveform and the physical accidentals alone. The in-app validation table marks each anchor accordingly — calibrated entries as “by construction”, predictive entries as “predicted (approx)” — and checks rate/OD/width anchors, energy bookkeeping, phase matching, and the velocity convergence flag.

15 Numerical architecture (brief)

The physics dictates the architecture, so a short note suffices:

- **One template, shifted.** $\mathcal{L}_0(\Delta_{\text{eff}}) = \mathcal{L}_0(0) - \Delta_{\text{eff}} S_v$ (Eq. (15)) lets every velocity share one Liouvillian and be assembled by cheap matrix shifts.
- **Δ -independent, per- T table.** $\bar{\chi}(\delta, \Delta_{\text{eff}})$ is computed on the needed grid; $\Delta \rightarrow$ sampled columns or interpolation coordinates, $T \rightarrow$ Maxwell weights *plus* the collisional dissipator (Sec. 4.3). The (Δ, T) map therefore builds one exact-union master grid per temperature and reuses it across all Δ .
- **Compiled OBE kernels.** `gabes/kernels.py` provides optional Numba kernels for the hot solve grids. FWM uses `floquet_chi_grid`; the absorption/Rydberg family uses the affine-scan kernel. The NumPy paths remain the reference fallback when Numba is unavailable.

- **Closed-form 2×2 .** The transfer matrix exponential is analytic (Appendix B), avoiding a per-point `expm`; its exponent arguments are overflow-clamped so runaway linear gain degrades to huge-but-finite values instead of NaN-poisoning the curve (Sec. 9).
- **Thread hygiene.** The small dense systems are kept out of oversubscribed BLAS threading with `core.blas_single_thread`; compiled kernels parallelize at the grid level, and `full_spectrum` computes the two Raman branches independently.
- **Headless readout.** Every scheme’s `observables` accepts `include_figures=False`, returning metrics/tables without building Matplotlib figures — the batch scanners and reports use this path (figure construction, not the OBE solve, dominates a warmed interactive call).

16 Validation, scope, and what is *not* modelled

Validated / first-principles. The direct seeded G_s/S_{dB} pipeline is regression-locked bit-for-bit, and at Ultra fidelity reproduces the Sim *et al.* measured operating point ($G_s \approx 14$, -7.8 dB) with the single residual $\ell_s = 0.74$. The optional Numba kernels are pinned to the NumPy reference paths, and the app-facing phase-detail modes add propagation physics without changing the atomic OBE definition. The genuine ^{85}Rb D_1 hyperfine CG factors (10, 35, 35, 28)/81 drive the coherent coupling, the polarization readout, and the spontaneous-emission branching; the macroscopic scale chain — CRC density, $p_F/[2(2I + 1)]$ normalization, self-broadening β , and the hardened arm/pump absorption channels — is tied to the AutoOD-validated absorption path ($< 0.1\%$ against measured cell transmission). The ideal intensity-difference noise is the conserved-difference result Eq. (38), and the collisional dissipator gives the observed interior temperature optimum.

Lumped / calibrated residuals. What still carries a knob: the dimensionless line-strength residual $\ell_s = 0.74$ (m -sublevel participation / beam geometry, not yet derived); the pump-scatter leakage fraction $\kappa = 0.1$ of the hardened Ultra noise (PBS extinction / spatial filtering, awaiting measurement); the biphoton absolute pair rate and (in the Calibrated model) the $g^{(2)}$ noise floor, which are reference-anchored; and the predictive biphoton *absolute* widths, which are approximate (telecom ≈ 0.50 vs $0.56(4)$ ns; Cs within $\sim 30\%$) pending the per-source calibration deferred in `docs/checklist.json`.

Ultra status and remaining future physics.

- Ultra includes single-pass refractive phase matching, Gaussian beam-overlap weighting, 64-segment dynamic pump-budget depletion, in-cell beam-splitter loss before detector efficiency, and — intrinsic since v5 — the hardened excess-noise model (arm-resolved linear absorption, OD-anchored pump scatter). `excess_noise_model=False` reproduces pre-hardening Ultra for A/B.
- Remaining zero-default technical noise channels (seed excess noise, EOM residual multimode noise) await calibrated measurements.
- Full Zeeman (m_F) substructure is available as a 24-level ^{85}Rb D_1 diagnostic manifold whose CG sums reproduce the lumped FWM strengths (consistency $\equiv 1.0$, reported but not multiplied in). The full 24-level Floquet scan ($M = 576$ vs $M = 16$) remains future work.
- Optical-pumping depletion of the $F = 3$ ground manifold when the pump parks near its resonances — the leading candidate for the experiment’s $+0.9$ GHz preference over the model’s shallow -2.0 GHz optimum — is not representable in the lumped steady-state model (Sec. 12).

- Full quantum-Langevin propagation of loss/noise (beyond the beamsplitter treatment) remains future work; the twin-beam photon statistics stay ideal-parametric estimates.

A Symbol glossary

symbol	meaning
Δ	one-photon (pump) detuning from $F = 2 \rightarrow F' = 3$
δ	two-photon (Raman) detuning
$\Delta_{\text{eff}} = \Delta - kv$	effective one-photon detuning of a velocity class
Ω	Rabi frequency (drive strength)
$\nu_{\text{HF}} = 3.0357 \text{ GHz}$	ground hyperfine splitting
$\Omega_{\text{beat}} = \Omega_{\text{HF}} \mp \delta$	ground-coherence beat (Floquet) frequency
$\Gamma/2\pi = 5.746 \text{ MHz}$	natural linewidth
$\gamma_{gg}(T, N)$	ground Raman dephasing: $2\pi \cdot 100 \text{ kHz floor} + N\sigma\bar{v}_{\text{rel}}$, Eq. (10)
$\gamma_{\text{opt}} = \beta N/2$	collisional optical dephasing (self-broadening), Eq. (9)
C_F^2	hyperfine Clebsch–Gordan line strength
$\bar{\chi}_{xy}$	dimensionless susceptibility (leg x , field y)
χ_{xy}	physical susceptibility, Eq. (33)
N	atomic number density (pure- ^{85}Rb CRC, Eq. (4))
T (matrix)	2×2 field transfer matrix $\exp(ML)$
G_s, G_c	seed / conjugate power gain
$S_{\text{ideal}} = (G_s - G_c)^2 / (G_s + G_c)$	ideal twin-beam noise, Eq. (38)
S, S_{dB}	intensity-difference squeezing (linear / dB)
$\eta = \text{QE}(1 - \text{loss})$	detection efficiency (0.8694 at reference)
$\varrho_{\text{norm}} = p_F / [2(2I + 1)]$	first-principles coupling norm (7/144 on branch $-$)
ℓ_s	residual line-strength calibration (<code>LINE_STRENGTH_FACTOR</code> ; app 0.74)
$\eta_{s,c}^{\text{arm}}$	hardened per-arm efficiencies, Eq. (45)
κ	pump-scatter leakage fraction (hardened Ultra, default 0.1)
Δk_z	seeded longitudinal phase mismatch
$\Delta k_z^{\text{abs}}, \Delta k_z^{\text{rel}}$	biphoton longitudinal absolute / calibrated relative mismatch
$\Delta k_x, \Delta k_{\text{vec}}$	biphoton transverse and vector phase mismatch
$\Phi_{\text{vec}} = \text{sinc}^2(\Delta k_{\text{vec}} L/2)$	biphoton vector phase-matching weight
$\psi(\tau)$	biphoton temporal waveform

B Closed-form 2×2 matrix exponential

Write $A = s\mathbb{K} + q$ with $s = \frac{1}{2} \text{Tr } A$ so that q is traceless. By the Cayley–Hamilton theorem a traceless 2×2 matrix satisfies $q^2 = -\det(q)\mathbb{K} = c^2\mathbb{K}$ with $c = \sqrt{-\det q}$. Hence

$$e^{qL} = \cosh(cL)\mathbb{K} + \frac{\sinh(cL)}{c}q, \quad e^{AL} = e^{sL} \left[\cosh(cL)\mathbb{K} + \frac{\sinh(cL)}{c}q \right]. \quad (58)$$

This is evaluated entrywise and batched over all detuning points in `core.matrix_exp_2x2`; the $c \rightarrow 0$ limit (where $\sinh(cL)/c \rightarrow L$) is handled explicitly to avoid the removable singularity. For the propagation matrix M of Eq. (30) this gives the transfer matrix $T = e^{ML}$ exactly, with no series truncation. The real parts of cL and sL are clipped at ± 350 before $\cosh / \sinh / \exp$ (two factors multiply; float64 overflows near 709): an unphysical runaway linear gain then saturates finite and is capped by the Manley–Rowe stage, instead of overflowing to `inf` and emerging as NaN from $\infty/(1 + \infty)$ in the saturation formula.

C File / function map

physics	function
Rabi from power	<code>rabi_freq</code>
density $N(T)$ (seeded, pure- ⁸⁵ Rb CRC)	<code>number_density</code>
density (other schemes / biphoton)	<code>rb85_density, number_density</code>
CG line strengths	<code>CF2, line_strength, cf2</code>
coupling norm $p_F/[2(2I + 1)]$	<code>physical_coupling_norm</code>
level model + Lindblad	<code>AtomModel, double_lambda_rb85</code>
collisional atom (per T)	<code>collisional_atom</code>
ground Raman rate $\gamma_{gg}(T, N)$	<code>ground_coherence_dephasing</code>
self-broadening $\Gamma + \beta N$	<code>self_broadened_gamma</code>
commutator super-op	<code>comm_super</code>
Liouvillian	<code>build_liouvillian</code>
Hamiltonians	<code>static_hamiltonian_at_DeFF_zero, sideband_hamiltonian</code>
Floquet steady state	<code>floquet_solve</code>
$\bar{\chi}$ matrix table	<code>chi_matrix_table, floquet_chi_grid</code>
velocity grid / Δ_{eff}	<code>velocity_grid, build_Delta_eff_axis</code>
Doppler average	<code>doppler_average</code>
Maxwell–Bloch gain / segmented propagation	<code>gain_from_chi</code>
2×2 matrix exp (overflow-clamped)	<code>matrix_exp_2x2</code>
pump-depletion cap	<code>pump_depletion_saturation</code>
ideal twin-beam noise	<code>ideal_twin_beam_noise</code>
squeezing (symmetric η)	<code>intensity_difference_squeezing_dB</code>
balanced two-arm noise	<code>balanced_twin_beam_noise</code>
in-cell loss + technical noise	<code>segmented_loss_noise_squeezing_dB</code>
arm linear OD (hardened)	<code>_arm_linear_od</code>
pump-scatter noise (hardened)	<code>_pump_scatter_noise</code>
Ultra segmented propagation	<code>_ultra_segmented_gain</code>
twin-beam statistics	<code>coincidence_stats</code>
full seeded pipeline	<code>compute_spectrum</code>
biphoton source	<code>GenericFWMSolver.compute_biphoton</code>
predictive biphoton waveform	<code>_predictive_waveform, _converged_predictive_waveform</code>
seeded Δk_z	<code>seeded_phase_mismatch_z</code>
phase matching	<code>phase_mismatch_vector, phase_mismatch_vector_grid, phase_matching_weight</code>
biphoton readout	<code>biphoton_stats</code>

Reference: G. Sim, H. Kim, H. S. Moon, *Sci. Rep.* **15**, 7727 (2025); Rb data from D. A. Steck, “Rubidium 85 D Line Data.”